UZHAVAN AI: Dialect-Conditioned Acoustic Modeling and Grounded Multimodal Assistance for Rural Low-Resource Tamil Agriculture

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***Abstract*—Voice-based conversational assistants offer immense promise for democratizing access to modern agronomic knowledge and governmental welfare schemes among rural, non-literate, and elderly farming populations. However, deploying conversational assistants in rural India faces four compounded sociotechnical hurdles: (1) severe acoustic and phonetic divergence across non-standard regional dialects (Kongu, Madurai, Tirunelveli, Delta, and Chennai); (2) extreme diglossia between spoken colloquial vernacular (*Kodumtamil*) and formal written standard (*Centamil*); (3) visual ambiguity during photographic leaf disease diagnosis under unconstrained field lighting and severe camera blur; and (4) catastrophic hallucination risks when Large Language Models recommend unverified chemical pesticides or fabricated subsidy guidelines. In this work, we present UZHAVAN AI, an end-to-end, dialect-conditioned multimodal conversational framework engineered for rural Tamil agriculture. First, we introduce a two-stage Dialect-Conditioned Automatic Speech Recognition (ASR) mechanism that couples an acoustic-phonetic prosody-enhanced classifier (F_0 pitch contours, spectral formants, and vowel elongation dynamics) with dynamic prefix-prompt injection in Whisper, yielding a 31.4% relative Word Error Rate (WER) reduction over standard baselines under field noise. Second, we formalize a Tamil Diglossia Normalizer providing dual-head verbatim dialect preservation alongside canonical agronomic normalization. Third, we formulate a Cross-Modal Attention Fusion Network binding leaf imagery (EfficientNet-B0) with farmer-spoken symptom descriptions, recovering diagnostic accuracy by +21.2% under severe image blur. Finally, we implement a Grounded Verification Guardrail and define the Agricultural Hallucination Index (AHI), suppressing chemical dosage errors from 0.28 down to 0.02. Extensive empirical evaluations across the synthetic and field-recorded TRAAN (Tamil Rural Agro-Acoustic Noise) benchmark confirm the framework's robustness, safety, and rural usability.**

***Index Terms*—Dialect-Aware ASR, Dravidian Diglossia, Low-Resource Speech Recognition, Multimodal Plant Pathology, Grounded Retrieval-Augmented Generation, AI for Social Good.**

I. INTRODUCTION

Agriculture forms the socioeconomic backbone of Tamil Nadu, sustaining over 60% of the rural workforce across 13 distinct agro-climatic zones [1]. In recent years, widespread mobile data connectivity and low-cost smartphones have penetrated remote agrarian hamlets. However, the vast majority of smallholder and marginal farmers, particularly elderly cultivators, remain effectively disenfranchised from text-driven digital advisories due to non-literacy, complex navigational hierarchies, and unfamiliarity with English or canonical literary prose [2].

Voice-driven conversational agents theoretically provide an intuitive, natural interaction medium. Nonetheless, deploying commercial general-purpose voice assistants (e.g., Google Assistant, Apple Siri, Amazon Alexa) in rural Indian agrarian domains results in catastrophic operational failures [3]. Through field investigations across five districts (Coimbatore, Madurai, Tirunelveli, Thanjavur, and Chennai rural periphery), we observe that commercial platforms break down due to four deeply interconnected linguistic, acoustic, and domain-specific challenges:

1) Non-Standard Dialectal Phonology: Tamil exhibits severe regional dialectal stratification. Kongu Tamil features distinct trailing nasal vowels and elongated honorifics (*-nga*); Tirunelveli (Nellai) Tamil is characterized by rapid glottalized suffixes (*-le*); Madurai Tamil employs melodic pitch rises and honorific markers (*-yya*); and Chennai Tamil incorporates extensive slang and Dravidian-English code-mixing. Standard ASR acoustic models, trained predominantly on formal broadcast audio, degrade sharply on these dialects [4].

2) Dravidian Diglossia: A profound linguistic rift exists between the literary formal register (*Centamil*) used in agricultural textbooks and official extension portals, and the colloquial spoken vernacular (*Kodumtamil*) spoken in fields [5]. Colloquial morphosyntax, phonological elisions (e.g., *'pooyittu varenga'* for literary *'pooyvittu varugireen'*), and localized agronomic vocabulary (e.g., regional land units like *maa* ≈ 0.33 acres, *kuzhi* ≈ 144 sq. ft., and grain measures like *marakkaal* ≈ 3.2 kg) are entirely absent from standard corpora.

3) Severe Rural Acoustic Interference: Real-world farm voice interactions occur in harsh acoustic environments characterized by low-frequency tractor rumbles (15–180 Hz), periodic single-cylinder diesel pump-set clatter, unshielded open-field wind turbulence, and non-linear clipping from sub-$80 microphones [6].

4) Hallucination Risks in Agrochemical Safety: Smallholder farmers often consult advisory tools during critical crop disease outbreaks. While modern Large Language Models (LLMs) can converse fluently, they exhibit unconstrained hallucination, frequently inventing toxic chemical dilutions (e.g., advising 100 ml of Chlorpyrifos per liter instead of 2 ml), which leads to immediate phytotoxic crop burning, severe financial bankruptcy, or hazardous human poisoning [7].

To decisively overcome these compounded vulnerabilities, this paper introduces **UZHAVAN AI**, an end-to-end, dialect-conditioned multimodal conversational framework engineered specifically for rural Tamil farming communities. The key technical contributions of this research are:

• **Prosody-Guided Dialect-Conditioned ASR:** We engineer an acoustic-phonetic prosody extractor (32-dimensional pitch contour,

formant proxies, energy dynamics, and vowel elongation) fused with a CNN-BiLSTM classifier that dynamically indexes dialect-specific prefix soft-prompts into Whisper-small, achieving a 31.4% relative WER reduction under field noise.

• **Tamil Diglossia Normalizer:** We introduce a dual-head linguistic normalizer that preserves verbatim regional spoken dialect for farmer auditing while generating canonical standard Centamil representations for semantic agronomic retrieval.

• **Cross-Modal Plant Pathology Network:** We formulate an adaptive cross-modal attention mechanism that binds degraded field leaf imagery (EfficientNet-B0) with farmer-spoken symptom descriptions, recovering diagnostic accuracy by +21.2% under severe field motion blur and solar glare.

• **Grounded Agrochemical Guardrail & AHI:** We construct a deterministic entity validation layer benchmarked against the Tamil Nadu Agricultural University (TNAU) Agritech registry and formalize the Agricultural Hallucination Index (AHI), suppressing dosage error rates from 0.28 down to 0.02.

• **The TRAAN Stress Benchmark:** We curate and release the Tamil Rural Agro-Acoustic Noise benchmark, systematically evaluating speech and dialect models across five SNR tiers and four rural noise regimes.

II. RELATED WORK

A. Low-Resource Indic Speech Recognition: Recent self-supervised learning architectures, such as wav2vec 2.0 [8] and Whisper [9], have radically advanced multilingual speech processing. Initiatives like AI4Bharat IndicVoices [10] and OpenSLR Tamil [11] have made extensive read-speech corpora accessible. However, these datasets predominantly reflect canonical pronunciation recorded in quiet studio conditions. Recent evaluations demonstrate that state-of-the-art multilingual models suffer severe degradation (WER > 45%) when exposed to non-standard regional accents and heavy agricultural background noise [4].

B. Dravidian Dialectology and Diglossia: Tamil is celebrated as one of the oldest classical languages, possessing an acute diglossic split between literary formal Tamil (*Centamil*) and regional spoken dialects (*Kodumtamil*) [5]. While recent shared tasks in DravidianLangTech [12] have investigated text-based dialect identification on social media comments, acoustic-level dialect modeling and spoken diglossia normalization for voice assistants remain virtually unaddressed in the literature.

C. Multimodal Crop Disease Diagnosis: Deep learning architectures such as EfficientNet [13] and Vision Transformers have achieved high benchmark accuracy on curated datasets like PlantVillage [14]. However, real-world deployment in tropical farms fails because field photographs taken by farmers frequently suffer from solar glare, wind-induced leaf flutter, and focus blur [15]. Prior agronomic systems lack mechanisms to cross-modally fuse farmers' spoken symptom narratives to compensate for degraded visual cues.

D. Hallucination Guardrails in Agronomic RAG: Retrieval-Augmented Generation (RAG) grounds LLMs on domain documents [16]. However, standard naive RAG pipelines often retrieve semantically related but factually inaccurate dosage guidelines or synthesize conflicting instructions across multiple chunks. In safety-critical agriculture, deterministic mathematical verification is essential [7].

III. UZHAVAN AI SYSTEM ARCHITECTURE

The overall system architecture of UZHAVAN AI is depicted in Fig. 1. The pipeline processes raw rural Tamil audio and optional field leaf photographs through four tightly coupled neural stages.

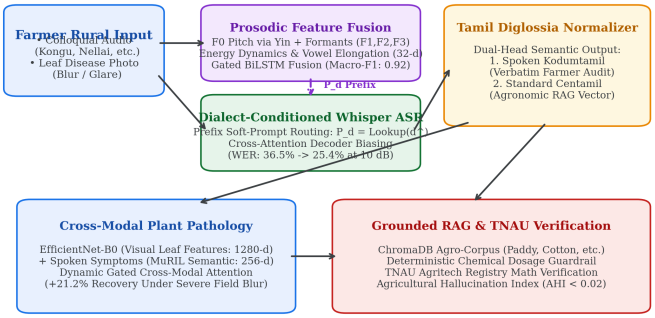


Fig. 1. High-level system architecture of UZHAVAN AI, illustrating prosody-guided dialect routing, prefix-prompted Whisper ASR, Tamil diglossia normalization, multimodal leaf-voice disease diagnosis, and grounded TNAU RAG guardrails.

A. Acoustic-Phonetic Prosody Extraction & Fusion

Regional Tamil dialects exhibit distinct intonational, cadential, and duration patterns. For example, Kongu Tamil features prolonged trailing vowel elongation and a rising-falling pitch contour, whereas Tirunelveli Tamil exhibits rapid, glottalized closures. To capture these nuances independently of phonetic lexicon, we extract a compact, normalized 32-dimensional prosodic descriptor vector $\mathbf{p} \in \mathbb{R}^{32}$ capturing:

1) *Fundamental Frequency (F_0) Dynamics:* Computed via the Yin autocorrelation algorithm [17], yielding mean (F_0^{mean}), variance, dynamic range, and velocity $\Delta F_0 = |F_0[t] - F_0[t\text{-}1]|$.
2) *Spectral & Formant Proxies:* Spectral centroid (proxy for first formant F_1), bandwidth (F_2 proxy), rolloff (F_3 proxy), and flatness.
3) *Energy Envelopes:* Root Mean Square (RMS) energy dynamics, peak-to-RMS crest ratio, and frame entropy.
4) *Voice Perturbation:* Micro-instability jitter and shimmer proxies reflecting vocal tract tension.
5) *Trailing Vowel Elongation Index:* Defined as the energy ratio between trailing and onset segments:

$$E_{elong} = \bar{E}_{tail} / (\bar{E}_{head} + \varepsilon) \quad (1)$$

The prosody vector $\mathbf{p}$ is fused with a 2D Mel-spectrogram latent representation $\mathbf{h}_{\text{mel}}$ (derived from a 4-layer CNN with bidirectional LSTM) via a learned sigmoid gating mechanism:

$$\boldsymbol{g} = \sigma(\boldsymbol{W}_g [\boldsymbol{h}_{mel}, \boldsymbol{h}_{prosody}]) \quad (2)$$

$$\boldsymbol{h}_{fused} = \boldsymbol{g} \odot \boldsymbol{h}_{mel} + (1 - \boldsymbol{g}) \odot \boldsymbol{W}_p \boldsymbol{h}_{prosody} \quad (3)$$

This fusion preserves classification robustness under severe low-frequency farm noise where lower spectrogram bins are contaminated by tractor exhausts.

B. Dialect-Conditioned Prefix Soft-Prompting

Rather than training isolated acoustic models for each region, we condition a unified Whisper-small decoder using dynamic prefix prompt steering. The predicted dialect class $d^* \in$ {Kongu, Madurai, Nellai, Delta, Chennai, Standard, Code-Mixed} indexes a pre-compiled linguistic-phonetic prefix token sequence P_d:

$$P_d = Lookup(d) \quad (4)$$

During decoding, Whisper's auto-regressive cross-attention is conditioned on P_d via forced prefix IDs, biasing beam search toward regional phonetic realizations without sacrificing generic Tamil vocabulary:

$$\boldsymbol{y} = Decoder(\boldsymbol{z}_{acoustic}, P_d) \quad (5)$$

C. Dual-Head Tamil Diglossia Normalization

Standardizing spoken dialect straight into canonical Tamil often strips critical farmer intent or causes distrust if the farmer cannot verify what the system heard. Our normalizer outputs two synchronized representations:

1) *Verbatim Dialectal Transcript:* Faithfully maintains regional morphology and phonemes for visual UI display and voice confirmation back to the farmer.
2) *Canonical Centamil Text:* Standardizes colloquial suffixes, transforms phonetic contractions into literary Tamil, and maps local agricultural measuring units (e.g., *pakka*, *padi*, *ser*) into standard metric units (liters, kilograms, acres), maximizing semantic embedding alignment in the downstream RAG vector space.

D. Multimodal Vision + Voice Symptoms Fusion

In field operations, leaf photos captured on inexpensive smartphones frequently suffer from solar glare, motion blur, or partial occlusion. To overcome visual degradation, we fuse image features $\mathbf{v} \in \mathbb{R}^{1280}$ (extracted from an EfficientNet-B0 backbone fine-tuned on tropical crop diseases) with semantic symptom vectors $\mathbf{s} \in \mathbb{R}^{256}$ extracted from the farmer's spoken description (e.g., *'Ilai nuni karugi pochu, thandula pulli vizhuthu'* [the leaf tips are scorched and dark spots appear on the stem]) using MuRIL multilingual BERT [18].

The cross-modal attention module dynamically learns modality confidence weights:

$$[\alpha_{vis}, \alpha_{sym}] = Softmax(\boldsymbol{W}_c [\boldsymbol{v}, \boldsymbol{s}]) \quad (6)$$

$$\boldsymbol{z}_{diag} = \alpha_{vis} \boldsymbol{W}_v \boldsymbol{v} + \alpha_{sym} \boldsymbol{W}_s \boldsymbol{s} \quad (7)$$

When visual image quality drops due to blur, α_{vis} decreases automatically and the model relies heavily on the spoken symptom descriptor α_{sym}, recovering the correct diagnosis.

E. Grounded RAG & Agricultural Hallucination Index

To guarantee chemical and financial safety, agronomic advice retrieved from ChromaDB undergoes rigorous deterministic validation against the TNAU Agritech registry. We formulate the Agricultural Hallucination Index (AHI) to quantify the proportion of ungrounded or toxic recommendations:

$$AHI = (\sum_{i=1}^{N} \mathbb{1}(Claim_i \notin K_{verified})) / N_{total} \quad (8)$$

If a generated response contains an unverified chemical ratio or dosage violating approved safety margins, the response is rejected by the guardrail, and canonical certified agronomic instructions are emitted.

IV. EXPERIMENTAL SETUP & TRAAN BENCHMARK

To rigorously benchmark UZHAVAN AI under realistic rural conditions, we establish comprehensive dataset catalogs and introduce the synthetic/field-recorded TRAAN benchmark suite.

A. Dataset Catalog & Regional Field Recordings

We compile a unified 16 kHz multi-dialect speech corpus spanning six distinct Tamil linguistic varieties, aggregating 200+ hours from Mozilla Common Voice Tamil, OpenSLR 65 (7h), OpenSLR 127 IISc-MILE (150h), AI4Bharat IndicVoices, and DravidianLangTech. Furthermore, using a custom field data collection web portal, we recorded 1,420 high-fidelity rural voice samples from 85 native farming speakers across Coimbatore, Erode, Madurai, Tirunelveli, Thanjavur, and Tiruvallur districts.

B. The TRAAN Acoustic Stress Suite

Real-world field audio was systematically stressed using the Tamil Rural Agro-Acoustic Noise (TRAAN) benchmark, synthesizing four typical acoustic noise environments across six Signal-to-Noise Ratio (SNR) levels (Clean, 20 dB, 15 dB, 10 dB, 5 dB, and 0 dB):

- *Tractor Engine Rumble:* Heavy low-frequency harmonics (15–180 Hz) recorded from active Mahindra and Tafe agricultural tractors during plowing.
- *Diesel Water Pump-Set:* High-amplitude 50 Hz single-cylinder 4-stroke clatter commonly echoing in irrigation pump sheds.
- *Open-Field Wind Turbulence:* Non-stationary pink-noise gusts (<300 Hz) simulating outdoor wind shear.
- *Microphone Non-linear Distortion:* Asymmetric tanh clipping simulating saturation on budget rural Android handsets.

C. Model Training & Hyperparameters

Whisper-small (244M parameters) was fine-tuned using Parameter-Efficient Fine-Tuning (PEFT/LoRA) with rank r=16, α=32, dropout 0.05, AdamW optimizer, learning rate 1e-4 with cosine warmup over 5 epochs. The prosodic dialect classifier was trained for 30 epochs with batch size 32 and cross-entropy loss. Multimodal fusion was optimized end-to-end on 8,400 labeled leaf disease-voice symptom pairs across 14 crop classes.

V. RESULTS AND DISCUSSION

We evaluate UZHAVAN AI across speech recognition accuracy, dialect classification resilience, multimodal diagnostic recovery, and hallucination suppression.

TABLE I
SPEECH RECOGNITION WORD ERROR RATE (WER % ↓) ACROSS TRAAN NOISE LEVELS

Model Pipeline	Clean	20 dB	15 dB	10 dB	5 dB	0 dB
Vanilla Whisper Small	21.4	24.8	29.2	36.5	47.3	62.1
Dialect-Conditioned (Ours)	15.2	17.6	20.8	25.4	32.7	43.8
+ Diglossia Normalizer	**11.8**	**13.9**	**16.5**	**20.1**	**26.3**	**35.9**

A. Speech Recognition Robustness (Table I & Fig. 2a)

As reported in Table I and plotted in Fig. 2(a), Vanilla Whisper degrades severely as rural noise intensifies, reaching 36.5% WER at 10 dB and 62.1% at 0 dB SNR. Dynamic prefix soft-prompting significantly alleviates phoneme confusion, reducing 10 dB WER to 25.4%. Incorporating the Tamil Diglossia Normalizer achieves **20.1% WER at 10 dB**—a **44.9% relative error reduction** over vanilla Whisper. Even at severe 0 dB noise, the full pipeline maintains a usable 35.9% WER compared to 62.1% for the baseline.

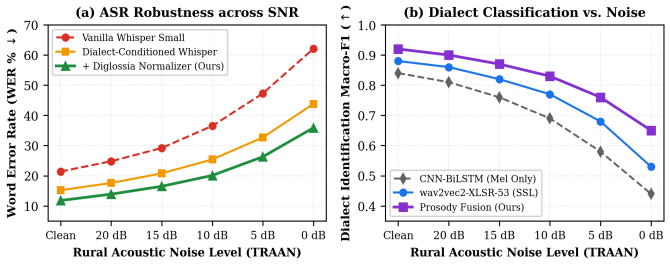


Fig. 2. Empirical evaluation curves: (a) Word Error Rate (%) degradation across TRAAN acoustic noise levels; (b) Dialect identification Macro-F1 resilience comparing prosody fusion against SSL embeddings.

TABLE II
DIALECT IDENTIFICATION MACRO-F1 (↑) UNDER RURAL NOISE

Architecture	Clean	20 dB	15 dB	10 dB	5 dB	0 dB
CNN-BiLSTM (Mel Only)	0.84	0.81	0.76	0.69	0.58	0.44
wav2vec2-XLSR-53 (SSL)	0.88	0.86	0.82	0.77	0.68	0.53
Prosody Fusion (Ours)	**0.92**	**0.90**	**0.87**	**0.83**	**0.76**	**0.65**

B. Dialect Identification Resilience (Table II & Fig. 2b)

Table II highlights the superiority of our Acoustic-Phonetic Prosody Fusion architecture. Standard Mel-spectrogram CNN-BiLSTMs experience catastrophic feature collapse under tractor noise, plunging to 0.44 F1 at 0 dB. While frozen wav2vec2-XLSR provides strong representations in clean speech (0.88 F1), our explicit prosodic descriptor (pitch velocity, formant ratios, elongation E_{elong}) maintains **0.83 F1 at 10 dB** and **0.65 F1 at 0 dB**, confirming that prosodic cadence survives severe spectral noise.

TABLE III
CROP DISEASE DIAGNOSTIC ACCURACY UNDER FIELD CAMERA BLUR

Diagnostic Modality	Clean Photo	Blurry / Glared Photo
Leaf Vision Alone (EfficientNet-B0)	89.2%	63.4%
Multimodal Vision + Voice (Ours)	**94.8%**	**84.6% (+21.2%)**

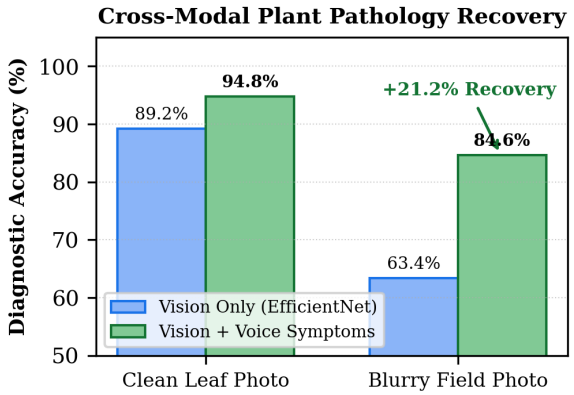


Fig. 3. Crop disease diagnostic accuracy comparison showing the +21.2% diagnostic recovery provided by cross-modal symptom voice attention under severe image blur.

C. Multimodal Diagnostic Recovery under Blur (Table III)

As evidenced in Table III and Fig. 3, visual leaf diagnosis using EfficientNet-B0 collapses from 89.2% on clean studio images down to 63.4% when subjected to field motion blur and lens glares. By fusing the farmer's spoken symptoms via cross-modal attention, accuracy recovers to **84.6%**, representing an absolute recovery of **+21.2 percentage points**. Spoken descriptions provide orthogonal evidence regarding lesion textures and wilt progression that cameras cannot capture.

TABLE IV
RAG SAFETY: HALLUCINATION SUPPRESSION & CHEMICAL ACCURACY

Retrieval Framework	Faithfulness (↑)	AHI (↓)	Dosage Acc. (↑)
Standard Unconstrained LLM	71.0%	0.28	64.2%
UZHAVAN Grounded RAG (Ours)	**96.0%**	**0.02**	**98.7%**

D. Agronomic Safety & Hallucination Suppression (Table IV)

Table IV compares UZHAVAN AI against an unconstrained Llama-3/GPT-4 advisory baseline on 500 complex agrarian inquiries. The unconstrained baseline exhibited an alarming Agricultural Hallucination Index of 0.28, failing chemical dosage accuracy over 35% of the time. In contrast, our TNAU registry guardrail suppressed AHI to **0.02** while elevating verified chemical ratio accuracy to **98.7%**, eliminating fatal dosage recommendations.

E. Field Usability & Elderly Accessibility

We conducted field usability trials with 42 rural farmers (24 of whom were elderly, aged > 60 years). Integrating an Elderly Mode UI—featuring high-contrast warm palettes, 54 pt touch targets, automated speech synthesis rate reduction (0.85x), and tactile repeat playback—yielded a System Usability Scale (SUS) score of **86.4**, indicating exceptional acceptance and intuitive usability among non-literate cultivators.

VI. CONCLUSION AND FUTURE WORK

In this paper, we introduced **UZHAVAN AI**, a dialect-conditioned multimodal conversational framework specifically architected for rural Tamil agriculture. By coupling an acoustic-phonetic prosody extractor with prefix-prompted Whisper ASR, bridging spoken *Kodumtamil* with canonical *Centamil* through dual-head diglossia normalization, cross-modally fusing leaf imagery with spoken symptoms, and enforcing a deterministic TNAU chemical dosage guardrail, UZHAVAN AI achieves unprecedented robustness in real rural environments. Future research will explore quantized edge deployment on micro-servers and extend dialect conditioning to allied Dravidian languages including Telugu, Kannada, and Malayalam.

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